PATRICK FROILAN S. DEL AYRE

Full-Stack Web, Native Android & Interactive Game Developer

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PROFESSIONAL PROFILE

4th-year BS Information Technology student (BSIT 4A-G1) at Bulacan State University specializing in end-to-end software engineering: modular React 19 web platforms, performant native Java Android applications, Node.js/Express REST APIs, geospatial telemetry with Leaflet GIS, and 120 FPS Unity 6 2D game architecture.

EDUCATIONAL BACKGROUND & MILESTONES

Bulacan State University (BulSU)

2022 - 2027 (4th Year Senior)

Bachelor of Science in Information Technology (BSIT 4A-G1)

- 4th Year (2026-2027 / Current): Capstone 2 full deployment (YTO Express Web Admin & Android Mobile Suite) & Hagonoy GroundWatch.
- 3rd Year (2025-2026): Systems Analysis & Design (SAD / Capstone 1) - system modeling, Bridge API sync, & CameraX prototype.
- 1st-2nd Year (2022-2025): Java OOP, relational database design (MySQL/PostgreSQL), web development fundamentals, and algorithms.

TECHNICAL CAPABILITIES

Languages:	Java (Android SDK), JavaScript (ES6+), C# (Unity 6), HTML5, CSS3, SQL
Frontend & GIS:	React 19, Vite, Tailwind CSS, Leaflet.js, MapLibre SDK, GeoJSON, Responsive WCAG UI
Backend & DB:	Node.js, Express, MongoDB, Mongoose, Server-Sent Events (SSE), JWT & OTP Authentication
Mobile & Games:	Android Studio, CameraX, ZXing QR, Unity 6 (2D Physics & Animation), Playwright E2E, Git

FEATURED ENGINEERING PROJECTS

YTO Express — Admin & Dispatch Web Portal (Capstone 1 & 2)

2025 - 2027 (Capstone 1 & 2)

React 19, Vite, Tailwind CSS, Node.js, Express, MongoDB, Leaflet GIS, SSE | github.com/Natoy0123/yto-express-web

- Central logistics command center with real-time KPI progress cards, rider rankings, and PDF/CSV/XLSX exports.
- Built a 10-hub Bulacan regional geofence monitor with Satellite/Geofence/Heatmap layers and live GPS alerts.
- Implemented bidirectional Bridge API sync (syncParcel, sendStatus, YTOQR1 QR protocol) with SSE live streaming.

YTO Express — Courier, Seller & Tracking Mobile App (Capstone 1 & 2)

2025 - 2027 (Capstone 1 & 2)

Java, Android SDK, MapLibre SDK, CameraX, ZXing QR, Material Design 3 | github.com/Natoy0123/YTO-Express-App

- Multi-role native Android app unifying Seller pickup booking, Customer GPS tracking, and Rider delivery execution.
- Integrated CameraX + ZXing high-speed waybill QR scanning and MapLibre turn-by-turn route polylines across hubs.
- Enforced 10m/20m doorstep GPS geofencing before unlocking Proof of Delivery (POD) photo capture and digital e-receipts.

Hagonoy GroundWatch — Civic Telemetry & Subsidence GIS Platform

2026

React 19, Vite, Tailwind CSS v4, Leaflet.js, GeoJSON, Playwright E2E | hagonoy-groundwatch-tik7.onrender.com

- Open-access civic GIS platform monitoring coastal land subsidence and tidal flooding across 26 Hagonoy barangays.
- Interactive Leaflet hazard map with polygon corridor filtering, Aquifer Stress Simulator, and citizen incident intake.

Slime Food Drop — 120 FPS Unity 2D Mobile Arcade Game

2026

Unity 6 (6000.3.0f1), C#, Android SDK, IL2CPP ARM64, 2D Physics | github.com/Natoy0123/SlimeFoodDrop

- Android portrait arcade game featuring Bloop the Slime, dynamic squash-and-stretch physics, and 10 handcrafted stages.
- Optimized touch input and rendering pipeline for 120 FPS mobile displays using Unity IL2CPP ARM64 native compilation.